`model_cory`: specification

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This documents `enightmodels::model_legacy()` as it is called on the model test page, with `model_legacy_config()` defaults. It is the production legacy model and the second of the two baselines the precinct × vote-mode model is compared against. Sections follow the same order as the main model's specification. The source is `enightmodels:::model_legacy_total` and `enightmodels:::model_legacy_cand`; this document describes those functions, not a reimplementation of them.

Configuration. The `model_legacy_config()` defaults in force:

`by_method`	`TRUE`	`min_unit_rep`	0.05
`strata`	`geostratum`	`tr_pre`	1
`stage_complete`	6	`tr_min`	0.25
`min_complete_per_stratum`	1	`tr_max`	4

1 Outcomes

The model runs in two stages. The first predicts **total votes** per geo × mode; the second predicts **each candidate's votes** using the first stage's reporting fractions. Both are ratio estimators against a past-election baseline `past`, computed within a stratum. The stratum is the `geostratum`, not the county.

Unlike the other two models, nothing is estimated on a share scale and nothing is modeled as a share: the model predicts candidate *vote counts* directly.

2 Turnout: total votes

Global pre-scale. Before anything else, the baseline is rescaled so that its statewide total matches what has been counted so far, floored at one:

$$t = \max\left(\frac{\sum_{\text{all cells}} \text{votes}}{\sum_{\text{all cells}} \texttt{past}},\ 1\right), \qquad \texttt{past} \leftarrow \texttt{past} \cdot t. \tag{1}$$

The floor at 1 means the baseline is never scaled *down*. Early in the night the counted total is small, so $t = 1$ and the step is inert; it becomes active only once counting is far enough along that the running total exceeds the baseline.

Completeness. Cells are aggregated to geo × mode × stratum, and a unit is complete when `stage` ≥ 6. This is a reported flag, not something inferred from the counts.

Stratum inflation. Strata containing at least one complete unit get a single inflation factor per mode,

$$\begin{aligned}\texttt{infl}_{\sigma m} &= \text{clamp}\left(\frac{\sum_{i\in\sigma}\texttt{past}_{im}\cdot(\text{votes}_{im}/\texttt{past}_{im})}{\sum_{i\in\sigma}\texttt{past}_{im}},\ 0.25,\ 4\right)\\ &= \text{clamp}\left(\frac{\sum_{i\in\sigma}\text{votes}_{im}}{\sum_{i\in\sigma}\texttt{past}_{im}},\ 0.25,\ 4\right),\end{aligned} \tag{2}$$

a `past`-weighted mean of the votes-to-baseline ratio. Strata with no complete unit take $\texttt{infl}=1$. Prediction then substitutes the count where a unit is complete and inflates the baseline where it is not:

$$\widehat{T}_{im} = \begin{cases}\text{votes}_{im} & \text{complete},\\ \texttt{past}_{im}\cdot\texttt{infl}_{\sigma(i)m} & \text{otherwise.}\end{cases} \tag{3}$$

Excess redistribution. A final per-geo pass reconciles predictions against counts that have overtaken them. With $x_{im}=\max(\text{votes}_{im}-\widehat{T}_{im},0)$ the overshoot in a cell, given 5% headroom when the unit is not complete,

$$\widehat{T}_{im}\leftarrow\max\left(\frac{\sum_{m'}\text{votes}_{im'}}{\sum_{m'}\widehat{T}_{im'}},\ 1\right)\widehat{T}_{im},\qquad \widehat{T}_{im}\leftarrow\widehat{T}_{im}+\begin{cases}x_{im} & x_{im}>0,\\ -\dfrac{\widehat{T}_{im}}{\sum_{x=0}\widehat{T}}\displaystyle\sum_{m'}x_{im'} & x_{im}=0.\end{cases} \tag{4}$$

Cells that have overshot are raised to their counted value; the cost is taken proportionally from the cells that have not. The geo total is conserved by construction. This is the model's one adding-up guarantee, and it operates on overshoot only — there is no corresponding treatment of cells that are short.

3 Candidates

The second stage uses the first stage's reporting fraction $\rho_{im}=\text{votes}_{im}/\widehat{T}_{im}$ to scale each candidate's past votes. Within a stratum × mode × candidate, over units that have reported at least $\texttt{min_unit_rep}=5\%$,

$$\texttt{ratio}_{\sigma mk}=\frac{\sum_i\text{votes}_{imk}/\rho_{im}}{\sum_i\texttt{past}_{imk}},\qquad \widehat{v}_{imk}=\begin{cases}\text{votes}_{imk} & \texttt{stage}\geq 6,\\ \texttt{past}_{imk}\cdot\texttt{ratio}_{\sigma mk} & \text{otherwise,}\end{cases} \tag{5}$$

with the fallback $\texttt{ratio}=\widehat{T}_{im}/\sum_k\texttt{past}_{imk}$ where a stratum has no qualifying unit. Dividing by ρ_{im} grosses each reporting unit up to its implied final count before the ratio is formed, so partially-counted units contribute on the same footing as complete ones.

4 Uncertainty

There is none. `model_legacy()` returns `pred_min` and `pred_max` as `NA`, and no draws, interval or variance are produced anywhere in either stage. On the model test page every `model_cory` interval column is consequently set to `NA`, and the model contributes a point estimate only.

Limitations

Stated plainly. Several of these are deliberate: this is a long-serving production model whose simplicity and predictability are part of its value.

- **No uncertainty at all.** The model cannot be assessed on coverage or interval width, and cannot contribute a probability of winning. Any comparison against it is necessarily a point-accuracy comparison.
- **No ceiling.** Nothing bounds predicted votes by registrants. The clamp in (2) bounds the *ratio* at 4× the baseline, which is a loose constraint and not a registrant-based one.
- **Candidate predictions need not sum to the turnout prediction.** The two stages are separate ratio estimators, and (5) applies a per-candidate factor with no constraint tying $\sum_k \widehat{v}_{imk}$ to $\widehat{T}_{im}$.
- **The inflation factor is computed over all units in a qualifying stratum.** A stratum qualifies on having one complete unit, but (2) then averages over every unit in it, including barely-started ones whose votes-to-baseline ratio is near zero. This biases `infl` *downward* early in the night, when few units are complete — the opposite direction to the usual partial-count failure.
- **Clamping rather than flagging.** $\text{clamp}(\cdot, 0.25, 4)$ in (2) silently transforms an out-of-range ratio. A ratio outside $[0.25, 4]$ is a strong signal that the baseline is wrong for that stratum, or that the feed is misaligned; clamping discards that signal rather than surfacing it.
- **Completeness is load-bearing.** Both stages branch on `stage` ≥ 6, so the model depends on a feed-supplied flag that in production arrives late, uses different vocabulary per state, and can move backwards. There is no fallback that degrades gracefully if the flag is wrong.
- **The stratum is the geostratum, not the county.** Dependence between precincts in the same county is not addressed, and no shrinkage is applied to stratum effects: (2) is an unpenalized stratum mean.